

Session 4: Operators and ExpressionsTopics:Arithmetic, Logical, Relational, Increment/Decrement.Operator precedence.Example: Simple calculator for addition, subtraction, etc.

Tasks

1.

Create a simple JavaScript function called calculateTotal that takes two numbers: itemPrice and quantity, and returns the total bill amount using arithmetic operators.

```
#include <stdio.h>
```

```
float calculateTotal(float itemPrice, int quantity)
```

```
{  
    return itemPrice * quantity;  
}
```

```
int main()
```

```
{  
    float itemPrice = 250.50;  
    int quantity = 3;  
    float total;  
  
    total = calculateTotal(itemPrice, quantity);  
  
    printf("Item Price: %.2f\n", itemPrice);  
    printf("Quantity: %d\n", quantity);  
    printf("Total Bill: %.2f\n", total);  
  
    return 0;  
}
```

```
C:\Users\gajja\OneDrive\Desk × + v
Item Price: 250.50
Quantity: 3
Total Bill: 751.50

-----
Process exited after 0.07137 seconds with return value 0
Press any key to continue . . . |
```

2.

Build a Flipkart-style discount calculator: given product price, discount percentage, and a boolean isMember, use arithmetic and logical operators to calculate the final price (apply an extra 5% off if isMember is true).

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    float price = 2000.00;
```

```
    float discount = 10.0;
```

```
    float discountAmount;
```

```
    float finalPrice;
```

```
    int isMember = 1;
```

```
    // Calculate normal discount
```

```
    discountAmount = price * discount / 100;
```

```
    finalPrice = price - discountAmount;
```

```
    // Extra 5% discount for member
```

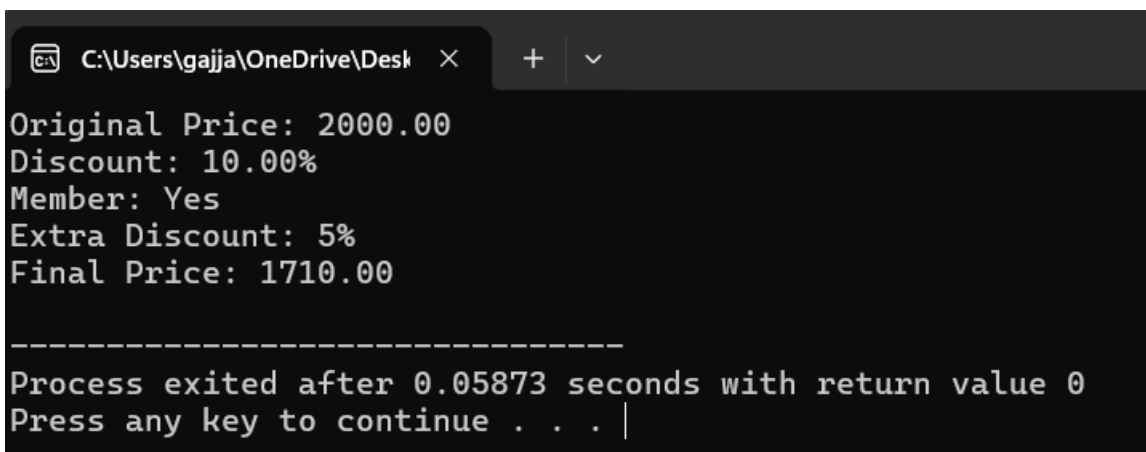
```
    if (isMember == 1)
```

```
    {
```

```
        finalPrice = finalPrice - (finalPrice * 5 / 100);
```

```
    }
```

```
printf("Original Price: %.2f\n", price);  
printf("Discount: %.2f%%\n", discount);  
  
if (isMember == 1)  
{  
    printf("Member: Yes\n");  
    printf("Extra Discount: 5%%\n");  
}  
else  
{  
    printf("Member: No\n");  
}  
  
printf("Final Price: %.2f\n", finalPrice);  
  
return 0;  
}
```



```
C:\Users\gajja\OneDrive\Desktop > .\program.exe  
Original Price: 2000.00  
Discount: 10.00%  
Member: Yes  
Extra Discount: 5%  
Final Price: 1710.00  
  
-----  
Process exited after 0.05873 seconds with return value 0  
Press any key to continue . . . |
```

3.

Write a function `isEligibleForOffer` that takes a user's age and total order value, and returns true if the user is 18 or older AND the order value is above 500, otherwise false.

Hint: Use relational and logical operators together.

```
#include <stdio.h>
```

```
int isEligibleForOffer(int age, float orderValue)
```

```
{
    if (age >= 18 && orderValue > 500)
    {
        return 1;
    }
    else
    {
        return 0;
    }
}
```

```
int main()
```

```
{
    int age = 20;
    float orderValue = 750;

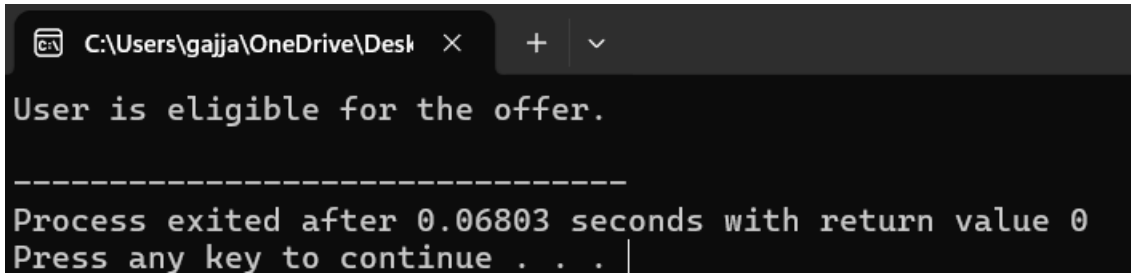
    if (isEligibleForOffer(age, orderValue))
    {
        printf("User is eligible for the offer.\n");
    }
    else
```

```

{
    printf("User is not eligible for the offer.\n");
}

return 0;
}

```



```

C:\Users\gajja\OneDrive\Desktop >
User is eligible for the offer.
-----
Process exited after 0.06803 seconds with return value 0
Press any key to continue . . . |

```

4.

Given three variables: likes, comments, and shares (all numbers), write code to check if a post is 'trending' on Instagram (at least 1000 likes OR more than 200 comments AND at least 50 shares). Print the result.

```
#include <stdio.h>
```

```

int main()
{
    int likes = 1200;
    int comments = 150;
    int shares = 60;

    if (likes >= 1000 || (comments > 200 && shares >= 50))
    {
        printf("The post is TRENDING.\n");
    }
    else
    {

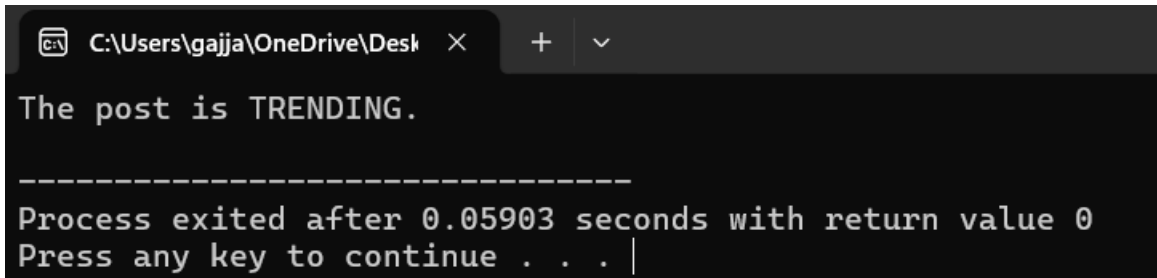
```

```

        printf("The post is NOT TRENDING.\n");
    }

    return 0;
}

```



```

C:\Users\gajja\OneDrive\Desktop
The post is TRENDING.
-----
Process exited after 0.05903 seconds with return value 0
Press any key to continue . . .

```

5.

Write a code snippet that demonstrates the difference between pre-increment (++count) and post-increment (count++) by logging the values before and after using both on a followerCount variable.

```
#include <stdio.h>
```

```

int main()
{
    int followerCount = 100;

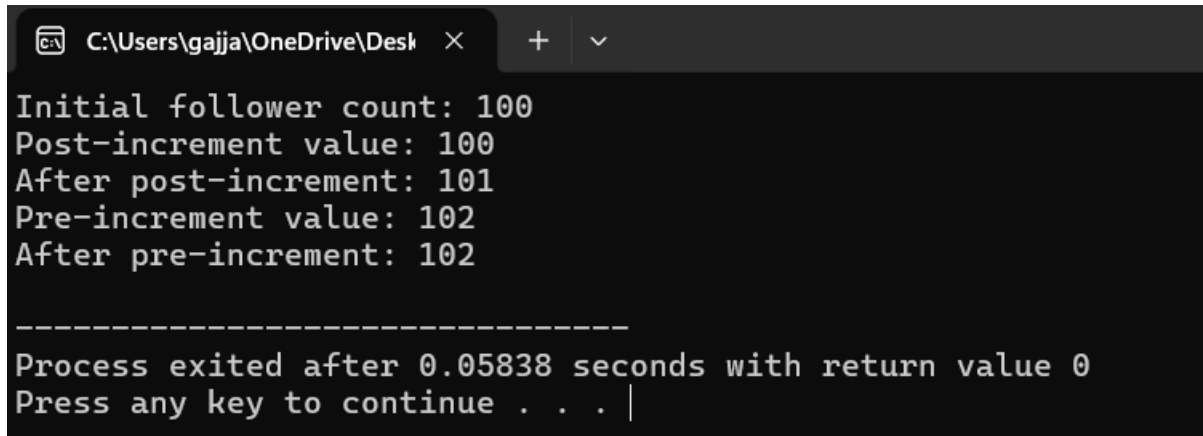
    printf("Initial follower count: %d\n", followerCount);

    // Post-increment
    printf("Post-increment value: %d\n", followerCount++);
    printf("After post-increment: %d\n", followerCount);

    // Pre-increment
    printf("Pre-increment value: %d\n", ++followerCount);
    printf("After pre-increment: %d\n", followerCount);
}

```

```
    return 0;  
}
```



The screenshot shows a Windows command prompt window with a dark background. The title bar at the top indicates the file path 'C:\Users\gajja\OneDrive\Desktop' and includes standard window controls. The output of the program is displayed in a monospaced font. It shows the initial follower count as 100, followed by post-increment and pre-increment operations and their respective values. A separator line of dashes is used to separate the output from the final exit message, which states the process exited after 0.05838 seconds with a return value of 0. The prompt 'Press any key to continue . . . |' is visible at the bottom.

```
C:\Users\gajja\OneDrive\Desktop >  
Initial follower count: 100  
Post-increment value: 100  
After post-increment: 101  
Pre-increment value: 102  
After pre-increment: 102  
  
-----  
Process exited after 0.05838 seconds with return value 0  
Press any key to continue . . . |
```